Do Now · retrieval

Recursion and algorithm tracing

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is an activation record?

Q2

When does code after recursive call execute?

Q3

What is auxiliary stack space for depth n?

Q4

Search for 12 in [2,4,6,8,10,12,14]. Midpoint indexes?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Trace recursive calls downward before calculating values on return.

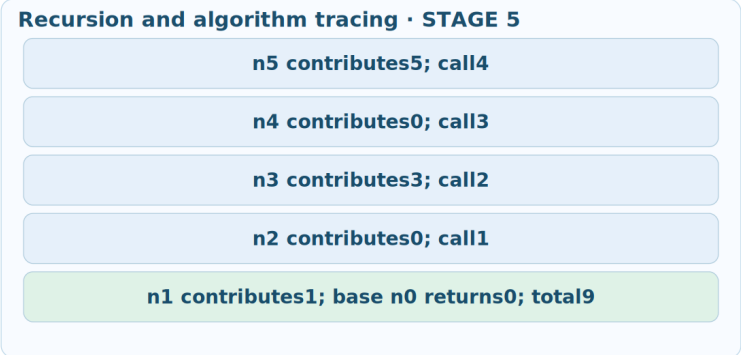
Core idea 2

Apply — Name the base case and variable that progresses toward it.

Core idea 3

Evaluate — Recursion depth consumes call-stack memory even when total time is linear.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

recursion — Subroutine calls itself.

base case — Stops additional calls.

call stack — Active subroutine frames.

unwinding — Returning from nested calls.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Predict output of printing before a recursive call for n=3.

Follow the worked method

1. Explain the result — Outputs 321 on descent. Nothing prints for n0.

Source model · final illustrated step

IDO · METHOD

Outputs 321 on descent. Nothing prints for n0.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the result
Outputs 123 on return as calls unwind.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace SumOdds(4), listing contributions and returned result. [6]

Question 2

For F(n): if n=0 return, else print(n) then F(n-1), state F(4) output and why. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Write an iterative C# method to sum the positive odd integers up to n (n ≥ 0), matching SumOdds(n). Compare its time and auxiliary space with the recursive method. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Printing after the recursive call appears on stack unwinding.

Use key vocabulary from memory

1. Explain recursion in your own words.

2. Explain base case in your own words.

3. Explain call stack in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.